Problem Set 9

1. Downsampling a finite sequence

Let $x[n]$ for $n = 0, \ldots, 11$ be $\{3, 1, -2, 4, 0, 5, 2, -1, 6, 3, 0, 2\}$. Downsample by $M = 3$.

(a) List $x_d[n]$.

(b) If the original sample rate is 12 kHz, what is the output sample rate?

(c) Which original indices are discarded?

2. Designing a decimator

You must decimate a sensor signal by $M = 4$. The useful input band extends to 0.18π and frequencies above 0.25π are known to be interference.

(a) Draw the block diagram.

(b) State the ideal aliasing boundary π/M.

(c) Propose reasonable FIR passband and stopband edges.

(d) Explain why a linear-phase FIR can be attractive and name one case where an IIR or minimum-phase filter might be preferable.

3. Strided convolution as a rate changer

Consider $x = [1, 2, 0, -1, 3, 1, 0]$ and a length-3 kernel $h = [1/4, 1/2, 1/4]$. Use the deep-learning cross-correlation convention (do not reverse h), no padding, and stride 2.

(a) Compute all stride-1 valid outputs.

(b) Keep every second result to obtain the stride-2 output.

(c) Explain why this is equivalent to filtering followed by downsampling.

(d) Is h a plausible anti-aliasing kernel? Justify qualitatively.

4. Upsampling and interpolation

Let $x[n] = \{1, 2, -1, 0\}$ and upsample by $L = 3$.

(a) Write the zero-inserted sequence.

(b) How many spectral images occur over one 2π DTFT interval?

(c) What ideal lowpass cutoff should the interpolation filter use?

(d) What DC gain should the filter have, and why?

5. Rational sample-rate conversion

Convert a stream from 30 kHz to 20 kHz.

(a) Reduce the rate ratio to L/M.

(b) Draw the conceptual $\uparrow L \rightarrow H(z) \rightarrow \downarrow M$ chain.

(c) Find the ideal combined-filter cutoff on the high-rate intermediate normalized-frequency axis.

(d) State the required DC gain of the interpolation/combined filter.

6. Transposed convolution and periodic artifacts (ECEN 5632 Problem)

A 1-D generator uses ConvTranspose1d with stride 2 and kernel length 5.

(a) Explain the zero-insertion + filtering interpretation.

(b) Why can kernel length 5 with stride 2 produce nonuniform overlap across output positions?

(c) Give two mitigation strategies.

(d) Explain how an anti-imaging-inspired initialization can help without guaranteeing artifact-free outputs.